Optimizing Offensive Strategies in Three-Robot Soccer through Hierarchical Tactical Planning and Multi-Robot Coordination

Author: LiMingzhe

Abstract

To address rule-handling flaws, a limited attacking approach, and insufficient coordinated support in the default strategy for Booster Champion three-on-three robot soccer, this paper proposes a hierarchical control method based on match-phase constraints, geometric decision-making, and short-horizon motion coordination. The system uses a fixed goalkeeper and dynamic division of duties between two field players, organizing attacks through role persistence, support-point search, and passing handovers based on ball displacement. Goal-line sampling and passing-lane clearance scoring select shooting targets. Goalkeeper ray blocking, conservative ball distribution, restart-specific organization, and A* navigation connect tactical intentions to control actions. Team-wide coordination checks motion conflicts jointly under short-horizon constant-velocity prediction. The improved method increases the win rate **from 40% to 50%, a 25% relative improvement over the baseline method**.

Keywords: robot soccer; multi-agent coordination; tactical planning; dynamic decision-making

I. Introduction

Three-robot soccer requires multiple agents to pursue the ball, attack, and defend together in a dynamic adversarial environment. The limited number of players creates a direct resource-allocation relationship between attacking support and goal protection. Repeated role switching, failure to respond to defensive positioning, and inappropriate ball-release actions can interrupt attacks or increase the risk of conceding.

Code and match-replay analysis of the Booster Champion default strategy shows that it mainly uses a division of duties with one attacker, one goalkeeper, and one covering defender. Attacks rely on the nearest player kicking directly toward the center of the opponent's goal, with little proactive passing or dynamic support. Its logic also contains flaws in retreating during opponent restarts, prolonged goalkeeper inactivity, and detecting actions that make no progress.

Based on this analysis, we construct a hierarchical control method around role allocation under match-phase constraints, geometric ball-release decisions, and multi-robot motion coordination. The main contributions are:

(1) Distinguishing basic logic flaws from tactical limitations and organizing baseline match evidence and before-and-after repair results to establish a starting point for further optimization.

(2) Using a fixed goalkeeper and dynamically assigned duties for two field players, together with role persistence, support-point search, ball-displacement-triggered passing handovers, and clearance-based shot selection, to organize continuous attacks; also designing ray blocking, conservative distribution, and restart-specific attacking and defensive organization.

(3) Connecting individual A* path planning with team-wide short-horizon motion coordination, checking the predicted minimum distance between robots before issuing commands, and defining independent match comparisons and module-ablation plans. The rules and thresholds in the method are manually specified, and their effects require experimental evaluation.

II. Methodology

A. Baseline Strategy Analysis

To establish the starting point for subsequent strategy optimization, we analyze the platform's default baseline strategy through code review, simulation runs, and match replays. Basic logic flaws are first identified and repaired, followed by analysis of tactical decisions and multi-robot coordination, separating basic execution problems from tactical optimization problems.

1) Basic Logic Flaws and Repairs. During observation and execution, we identified logic flaws in restart rule handling, goalkeeper state management, and detection of actions without progress. Table 1 presents representative match evidence before repair; all times refer to the remaining time in the corresponding match.

Table 1. Basic logic flaws in the baseline strategy and their repair context

Issue	Pre-repair match evidence	Cause and impact
Continued ball pursuit during opponent restarts	With 581.17 s remaining, robot1 was only 0.55 m from the ball versus the required minimum of 1.45 m; the player was sent off for 30 s and the opponent retook the goal kick.	_act_opp_set_play() directly executes normal attacking logic without a retreat action, failing to maintain the legal distance during an opponent restart.
Prolonged inactivity in front of goal	With 247.26 s remaining, robot3 was in front of our goal, 2.19 m from the ball, and received a 30 s penalty for inactivity.	The goalkeeper stays at the center of our penalty area after arriving, with no close-ball advance or repositioning mechanism after inactivity.
Continued waiting without progress	With 238.58 s remaining, the referee ruled that no player had touched the ball for 30 consecutive seconds and restarted with a dropped ball; the last goal kick by our team was first touched only 40.28 s after it was taken.	No detection of an action being executed while the ball remains stationary, and no timeout exit or re-approach mechanism, causing attacking stagnation.

We completed the corresponding logic repairs for these flaws. This stage targets anomalies in rule execution and action-state handling and does not involve tactical changes.

2) Post-Repair Test Results. Ten rounds of repair testing were conducted against Booster AI, and performance before and after repair was compared using the existing records in Table 2. The original baseline recorded 4 wins, 2 draws, and 4 losses, 62 penalties, and 2.4 goals conceded per match; after repair, the results were 3 wins, 4 draws, and 3 losses, 24 penalties, and 3.1 goals conceded per match.

Table 2. Test results before and after basic logic repair

Metric	Original baseline	Repaired baseline	Change
Wins/draws/losses	4/2/4	3/4/3	N/A
Total violations	62	24	61% decrease
Goals conceded per match	2.4	3.1	29% increase

According to Table 2, the win rate fell from 40% to 30%, a decrease of 10 percentage points; total violations fell from 62 to 24, a reduction of approximately 61.3%; average goals conceded rose from 2.4 to 3.1, an increase of approximately 29.2%.

3) Baseline Tactics and Their Limitations. We further examine the baseline strategy's tactical behavior. It assigns one attacker, one goalkeeper, and one covering defender, dynamically allocating roles according to player positions. Its main decision rules are listed in Table 3.

Table 3. Tactical rules of the baseline strategy

Aspect	Baseline tactic
Attacker selection	Select the available player nearest the ball; use an approximately 0.3 m switching buffer to reduce repeated role changes.
Attacking action	First move behind the ball, then approach and kick toward the center of the opposing goal.
Shot selection	Use default power 5 without seeking openings based on opponent positioning; proactive passing combinations are absent.
Goalkeeper selection	Choose the player nearest our goal from the remaining players.
Goalkeeping	Return to the center of our goal area and wait facing the ball, without actively advancing.
Defensive cover	The third player stands on the ball-to-own-goal line, approximately 3 m from the ball, forming a second defensive line.
Our kickoff	Lock in the nearest player and kick forward along a fixed, slightly offset direction; one remaining player guards the goal and the other waits in place.
Opponent kickoff	One player guards the goal while the other two retreat to fixed positions outside the center circle.
Restarts	The original strategy reuses normal-play logic for both teams' restarts, without dedicated attacking or defensive scripts; repairs of rule-handling flaws are shown in Table 1.

In attacking organization, the baseline mainly relies on one player approaching the ball and shooting directly toward the center of the goal, with little support or proactive passing. The shooting target and power are relatively fixed and do not adjust to defenders' positions; consequently, the same attacking direction may be selected repeatedly even when the center of the goal is blocked.

In multi-robot coordination, the goalkeeper and covering defender provide longitudinal protection, but their positions are relatively fixed and the remaining players rarely offer proactive attacking support. This division of duties may limit attacking width and continuous passing, and lacks joint decisions based on teammate positions, opponent pressure, and passing lanes.

In adaptation to game situations, kickoffs and restarts mainly depend on fixed actions or normal-play logic, without situation-specific tactical scripts. The subsequent method therefore uses a fixed goalkeeper and two cooperating field players, introducing passing handovers, clearance-based shot selection, restart-specific organization, and motion coordination. The competitive effects of these mechanisms will be tested in independent experiments.

B. Improved Strategy: Hierarchical Control and Multi-Robot Coordination

1) Overall Architecture and State Representation

For three-on-three robot soccer, we construct a hierarchical control method based on match-phase constraints, geometric decision-making, and short-horizon motion coordination. The system consists of match-phase management, team tactical decision-making, individual action control, and team-wide motion coordination. In each control cycle, it first determines the match phase from the referee state and filters actionable players; it then generates role assignments and passing, shooting, and positioning intentions, checks team-wide motion conflicts, and finally issues control commands. Figure 1 shows the overall workflow.

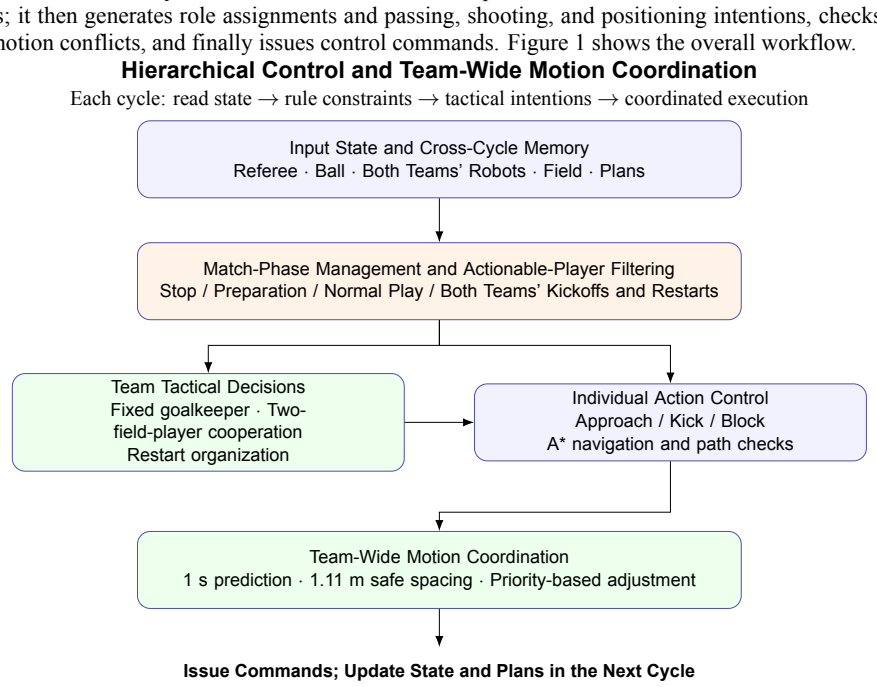


Figure 1. Hierarchical control workflow. Match-phase constraints precede tactical decisions, and team-wide coordination jointly handles motion conflicts before commands are issued.

A two-dimensional field coordinate system is used, with our goal in the negative x direction and the opponent's goal in the positive x direction. The input state at time t is represented as:

$$s_t = (g_t, \mathbf{b}_t, \mathcal{R}_t, \mathcal{O}_t, \mathcal{F}) .$$

Here, g_t is the referee state, b_t is the ball position, R_t and O_t are the state sets of our robots and opposing robots, respectively, and F contains field geometry parameters. Robot states include position, orientation, action mode, fallen status, and penalized status. The system additionally maintains cross-cycle memory for attacking roles, passing plans, restart plans, and shooting targets; this memory is not part of the instantaneous observation tuple above.

The referee state is mapped to seven phases: normal play, our kickoff, the opponent's kickoff, our restart, the opponent's restart, preparation, and stop. The stop phase cancels movement and kicking; preparation uses preset positions; opponent restarts allow only retreating and defensive positioning. Penalized players, players not yet ready, and players with unknown positions are excluded from role assignment. The current implementation uses a fixed attacking direction and does not include automatic coordinate transformation after switching ends.

2) Fixed Goalkeeper and Dynamic Field-Player Duties

Robot 1 is the fixed goalkeeper. The other two field players handle attacking and support duties during normal play. The initial primary attacker is the actionable field player closest to the ball:

$$i_t^* = \arg\min_{i \in \mathcal{A}_t} \|\mathbf{p}_i - \mathbf{b}_t\|_2,$$

Here, A_t is the set of actionable field players and p_i is player i's position. To reduce frequent switching, the newly selected primary attacker retains the role during the first 1.2 s as long as its distance to the ball does not exceed the nearest player's distance by more than 1.5 m; afterward, the persistence margin shrinks to 0.3 m. During pass reception, the intended receiver is preferentially retained until reception ends, times out, or the plan becomes invalid.

The nearest ball distances of the two teams, d_our and d_opp, indicate the opponent's proximity advantage. When both conditions below hold, the current attacking combination is canceled: one field player pursues and pressures the ball, while the other retreats to block between the ball and our goal:

$$d_{\text{opp}} < 0.8 \text{ m}, \qquad d_{\text{opp}} + 0.3 \text{ m} < d_{\text{our}}.$$

This criterion indicates only a geometric proximity advantage and is not confirmation of actual possession.

3) Two-Player Coordinated Attack and Passing Handover

Normal play uses four high-level actions: pursuing the ball, positioning, passing, and shooting. The primary attacker first decides whether to shoot based on shooting distance and lane obstruction; otherwise, it assesses pass feasibility, or adjusts its position behind the ball. The supporting player searches candidate points for combination space.

Six support candidates are generated relative to the ball by combining longitudinal offsets {1.5, 0, -1.5} m with lateral offsets {-2, 2} m, and are restricted to an area at least 0.65 m from the field boundary. The score for candidate point q is:

$$J_{\text{sup}}(\mathbf{q}) = 3I_{\text{clear}}(\mathbf{q}) + \min(d_{\text{opp}}(\mathbf{q}), 3) + 0.4\Delta x - 0.25\|\mathbf{p}_r - \mathbf{q}\|_2.$$

Here, I_clear indicates a clear passing lane, d_opp(q) is the distance from the candidate to the nearest visible opponent, Delta x is the longitudinal offset used to generate the candidate, and p_r is the receiver position; all distance values are in meters. The system selects the highest-scoring point, jointly considering the passing corridor, opponent distance, forward direction, and movement cost.

Lane checking uses the shortest distance from a robot's position to the ball-to-target segment: excluding the passer and receiver, any visible robot within 0.8 m of the segment blocks the lane. An ordinary pass requires the passer to be no more than 0.8 m from the ball, a passing distance of 1.5-4.0 m, and a target inside the field's safe region. Normally, the teammate must be at least 0.5 m ahead of the ball; when the nearest opponent is within 1.8 m of the ball, a backward pass to a teammate no more than 1.5 m behind the ball is allowed.

The passing target is fixed at the receiver's position when the plan is created. After a kick command is

submitted, ball positions continue to be observed rather than immediately declaring the pass complete. Let b_0 be the ball position before release and q the target. Role handover starts only when both conditions below hold:

$$\|\mathbf{b}_t - \mathbf{b}_0\|_2 \geq 0.25 \text{ m}, \qquad (\mathbf{b}_t - \mathbf{b}_0)^\top (\mathbf{q} - \mathbf{b}_0) > 0.$$

The original passer then moves into support and the receiver pursues the incoming ball. The plan is canceled if release waiting exceeds 3 s, the lane is blocked, the ball moves in the opposite direction, a participating player becomes unavailable, or the opponent gains a proximity advantage. After ball movement, reception lasts at most 5 s; normal decision-making resumes when the receiver is within 0.8 m of the ball or another player is clearly closer. This handover is triggered by position changes and is not confirmation of a successful pass or completed ball control.

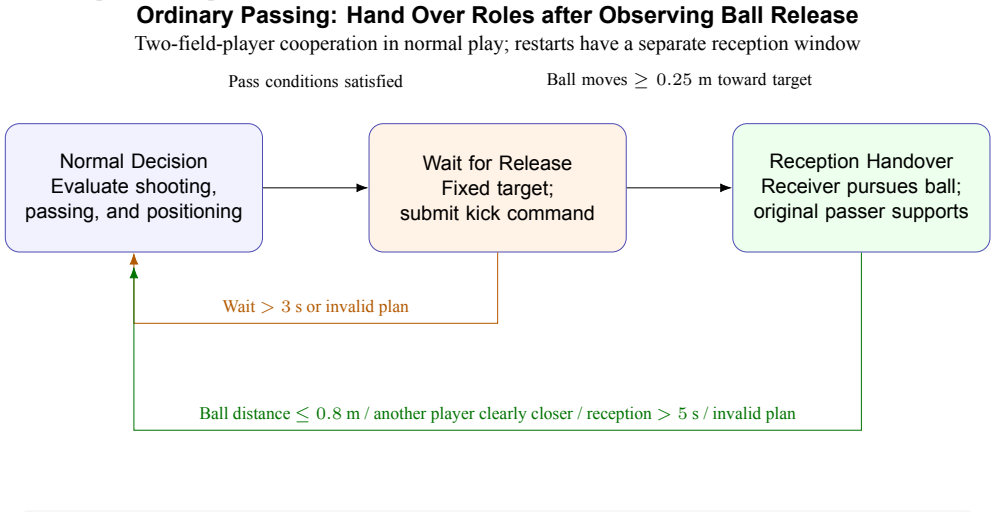


Figure 2. Handover states for an ordinary pass. Release waiting and post-movement reception are timed separately; invalidation returns control to normal decision-making.

4) Clearance-Based Shot Selection

The system uniformly samples 21 targets along the opponent's goal line, leaving a 0.25 m margin at each post. For candidate q_k=(x_g,y_k), it computes the minimum clearance of the ball-to-target segment relative to robots ahead:

$$C_k = \min\left\{1.2,\ \min_{j \in \mathcal{B}_t} \left[d\big(\mathbf{p}_j, \overline{\mathbf{b}_t \mathbf{q}_k}\big) - r_j\right]\right\}, \qquad J_{\text{shot}}(\mathbf{q}_k) = C_k - 0.08|y_k|.$$

B_t contains visible opponents and other teammates between the ball and the opposing goal line, and r_j is the configured radius of robot j; distance values are in meters. With no blockers, clearance takes the truncation ceiling of 1.2 m. The score favors greater lane clearance and slightly favors the goal center when clearances are similar. The previous cycle's target is retained if its score is within 0.08 of the current best score, reducing left-right switching.

During normal coordinated attack, shooting is invoked only if the ball-to-target distance is no more than 4.5 m and the lane passes an additional 0.8 m spacing check. While pursuing the ball, the approach target is set 0.35 m behind it along the intended release direction, aligning approach and kick directions.

Before execution, the kicking direction is transformed into the robot's body frame, and right-foot release-angle compensation is applied exactly once at the final control interface:

$$\theta_{\text{cmd}} = \text{wrap}(\theta_{\text{target}} - \theta_{\text{robot}} - \delta), \qquad \delta = 5^\circ.$$

The power parameter is 5 for an ordinary shot and 1 for a coordinated pass. These are execution-interface parameters and do not directly specify actual ball speed.

5) Goalkeeper Ray Blocking and Conservative Distribution

The goalkeeper switches among close-range distribution, collection inside the penalty area, ray blocking, and recovery positioning according to match constraints, ball position, and opponent threats. In normal play, close balls are handled first. If the ball is inside our penalty area, the goalkeeper actively approaches behind it and passes or clears once within close range.

A threatening opponent must satisfy all three conditions: it is less than 5 m from the ball; the ball is in the forward half-plane of its orientation; and the ray pointing from the opponent through the ball and beyond intersects our penalty area. Let o be the opponent position and b the ball position. The ray and nearest point are:

$$\mathbf{l}(\tau) = \mathbf{b} + \tau(\mathbf{b} - \mathbf{o}), \quad \tau \geq 0; \qquad \tau^* = \max\left(0, \frac{(\mathbf{p}_g - \mathbf{b})^\top (\mathbf{b} - \mathbf{o})}{\|\mathbf{b} - \mathbf{o}\|_2^2}\right), \quad \mathbf{q}_g = \mathbf{l}(\tau^*).$$

For the valid threatening opponent closest to the ball, the formula gives q_g, the closest point on the ray to goalkeeper position p_g, as the blocking target. Penalty-area intersection determines the threat, while the blocking target is computed on the entire ray; the navigation layer then handles boundaries and obstacles. If the opponent and ball coincide, the ray direction degenerates; its handling rule is [to be completed].

Close-range distribution uses entry and exit thresholds of 0.7 m and 0.95 m. Passes are made only to ready teammates with fresh observations who are positioned ahead, with checks on distance, turning angle, and lane risk; observation-freshness, distance, and turning-angle thresholds are [to be completed]. To conservatively estimate how close opponents may come to the lane during kick preparation and ball travel, the risk spacing is:

$$D_{\text{risk}}(\ell) = d_0 + v_o \left(T_{\text{prep}} + \frac{\ell}{v_b}\right), \quad d_0 = 0.8 \text{ m},\ T_{\text{prep}} = 2 \text{ s},\ v_o = 1.5 \text{ m/s},\ v_b = 3 \text{ m/s}.$$

Here, ell is the passing distance. A pass is accepted only if every opponent's distance to the passing segment exceeds this spacing. v_b is an assumed ball speed for risk estimation, distinct from the kick-power parameter. With unreliable observations or no feasible pass, a clearance direction is selected from forward 0 degrees, plus or minus 30 degrees, and plus or minus 60 degrees by considering lane space and turning cost. Field players handling close balls inside our goal area reuse this mechanism.

6) Attacking and Defending Restarts

Our restarts use three cooperative phases: preparation, release, and reception. The fixed goalkeeper stays back, the nearest actionable field player takes the restart, and the other receives. Candidate landing points in different directions are generated for kickoffs, corners, goal kicks, kick-ins, and free kicks. After obstructed lanes are excluded, the landing point minimizing the following expression is selected:

$$\mathbf{q}^* = \arg\min_{\mathbf{q} \in \mathcal{Q}_{\text{clear}}} \left[\|\mathbf{p}_r - \mathbf{q}\|_2 - 0.4 \min(d_{\text{opp}}(\mathbf{q}), 5)\right].$$

Q_clear is the set of candidates with clear lanes, and distances are in meters. The objective jointly considers the receiver's travel cost and space around the landing point. The receiver first runs toward the target while the taker waits approximately 0.8 m behind the ball. The ball is released when the receiver is within 0.1 m of the target and the lane is clear. After waiting more than 6 s, release toward a still-clear target is allowed; a timeout does not force a kick into a blocked lane.

After the ball is observed at least 0.25 m from its starting point, the taker stops touching it and moves diagonally forward into support, while the receiver actively approaches the incoming ball. Subsequent attack starts when the receiver is within 0.65 m of the ball. The plan may persist across the transition from a restart to normal play; after normal play begins, the reception window ends at most 8 s after ball movement starts. Penalties are handled separately: the taker shoots and the other field player stops moving.

Opponent restarts use type-specific defense. For direct free kicks in our half, field players form a wall along the direction from the ball to our goal center, by default 2 m from the ball with 0.6 m lateral spacing. Corners protect the near post and goalmouth. Kick-ins and indirect free kicks prioritize blocking lanes to potential receivers. Kickoffs, goal kicks, and distant free kicks use front-back layered positioning.

Defensive roles are retained during the same restart while the available-player set remains unchanged. If the fixed goalkeeper is unavailable, the actionable player nearest our goal temporarily protects it. Defense executes positioning only, while the path layer preserves the ball-exclusion region. Restrictions are released only when the referee phase ends, not automatically when the ball moves. Opponent penalties use a retreat-and-wait strategy.

7) Path Planning and Team-Wide Temporal Avoidance

Individual navigation represents the ball, opponents, and goal structures as circular obstacles, enabled according to action and referee phase. Targets are first restricted to the field's safe region; occupied targets are replaced by nearby available grid points. Global planning uses eight-connected A* at 0.35 m resolution, with path length as cost and Euclidean distance as heuristic. Both search edges and smoothing shortcuts are checked for intersection between the full segment and obstacles. If the start is already inside an inflated obstacle region, escape is allowed only in directions that do not deepen overlap and increase distance from the obstacle, handling disengagement near goalposts, nets, or ball-exclusion boundaries.

The team layer first gathers all motion intentions and predicts constant-velocity motion over the next 1 s. Each player is approximated as a disk of radius 0.48 m, with an additional 0.15 m error margin, giving a safe center-to-center spacing of:

$$D = 2 \times 0.48 + 0.15 = 1.11 \text{ m}.$$

For any two players, let r be relative position, v relative velocity, and H=1 s the prediction horizon. With nonzero relative velocity, the time of closest approach and acceptance condition are:

$$\tau^* = \text{clip}\left(-\frac{\mathbf{r}^\top \mathbf{v}}{\|\mathbf{v}\|_2^2}, 0, H\right), \qquad \|\mathbf{r} + \tau^* \mathbf{v}\|_2 \geq D.$$

If relative velocity is zero, the current distance is checked directly. Continuous-time closest-distance checking allows players to pass through the same location at different times and identifies head-on conflicts that discrete sampling may miss. Coordination priorities are the fixed goalkeeper, the field player nearest the ball, and other players, in that order. The system jointly searches candidate velocities such as original speed, reduced speed, waiting, and low-speed lateral movement, preserving higher-priority intentions where possible; specific velocity candidates and computational overhead are [to be completed]. Stopped or non-actionable players remain spatial occupancies in the checks.

The low-level kicking interface includes automatic approach, so kicking intentions participate in coordination through approximate trajectories toward the ball. If no acceptable joint solution exists, kicking is canceled and movement is stopped. Spacing checks in this short-horizon model do not guarantee collision-free motion throughout actual execution.

III. Experiments and Results

A. Experimental Setup

The experimental platform is Booster Champion and the opponent is Booster AI. The original default strategy and the complete proposed method are planned to be evaluated separately to distinguish basic repairs from tactical optimization. Each group is planned to play five matches as the home team and five as the away team, for a total of ten matches.

Simulation version, opponent version, match duration, field, robot configuration, and referee rules should remain identical across groups. Completed baseline-repair tests and planned full-method tests are assigned separate identifiers to avoid merging results from different batches.

The main metrics are wins/draws/losses, win rate, goals scored and conceded, goal difference, and our team's penalty count.

B. Experimental Results

Table 7. Match results of the original default strategy and the proposed method

Metric	Baseline	Ours
Win rate	**40%**	**50%**
Matches played	10	10
Wins	4	5
Draws	2	4
Losses	4	1
Total goals scored	23	28
Total goals conceded	24	25
Goal difference	-1	3
Total team penalties	62	28

IV. Conclusion

This paper analyzes limitations of the default three-robot soccer strategy in rule execution, attacking coordination, and adaptation to game situations, and develops a hierarchical control method combining match-phase constraints, geometric decision-making, and short-horizon motion coordination. With a fixed goalkeeper and dynamically assigned duties for two field players, the method integrates support-point selection, ball-displacement-triggered passing handover, clearance-based shooting, goalkeeper ray blocking and conservative distribution, restart-specific organization, and team-wide motion coordination into a unified workflow.

In the existing baseline-repair comparison, penalties fall from 62 to 24, but win rate falls from 40% to 30%, and average goals conceded rise from 2.4 to 3.1. This shows that reducing rule-execution anomalies does not necessarily produce competitive gains in the current sample.

With our proposed improved method, the match win rate increases **from 40% to 50%, a relative improvement of 25%**, effectively improving strategy performance.

V. Appendix

A. Hardware and Software Environment

Unless otherwise marked, cloud experiments use Alibaba Cloud hpc8ae high-performance computing-optimized instances. Local experiments marked with an asterisk (*) use a MacBook Pro.

Item	Cloud platform	Local platform
Computing platform	Alibaba Cloud hpc8ae	MacBook Pro
CPU	64 vCPU	Apple M3 Pro, 12 cores
Memory	256 GiB instance specification	36 GiB
Host operating system	Alibaba Cloud Linux 4 LTS, 64-bit	macOS 26.6 Beta, 25G5057c
Agent runtime	Official Docker container, ROS 2 Humble	Official Docker container, ROS 2 Humble
Agent Python	Python 3.10	Python 3.10
Simulation image	virtual-robot 0.9.4-alpha-humble	virtual-robot 0.9.4-alpha-humble
Build image	agent-dev 0.9.17-alpha-humble	agent-dev 0.9.17-alpha-humble
Execution per round	10 independent matches in parallel	1 match
Strategy computation	CPU computation	CPU computation

B. Simulation and Evaluation Settings

Simulation and evaluation use the platform's official default settings.